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INSTALLER GUIDE · BATTERY COMMUNICATION

Growatt: parallel battery communication

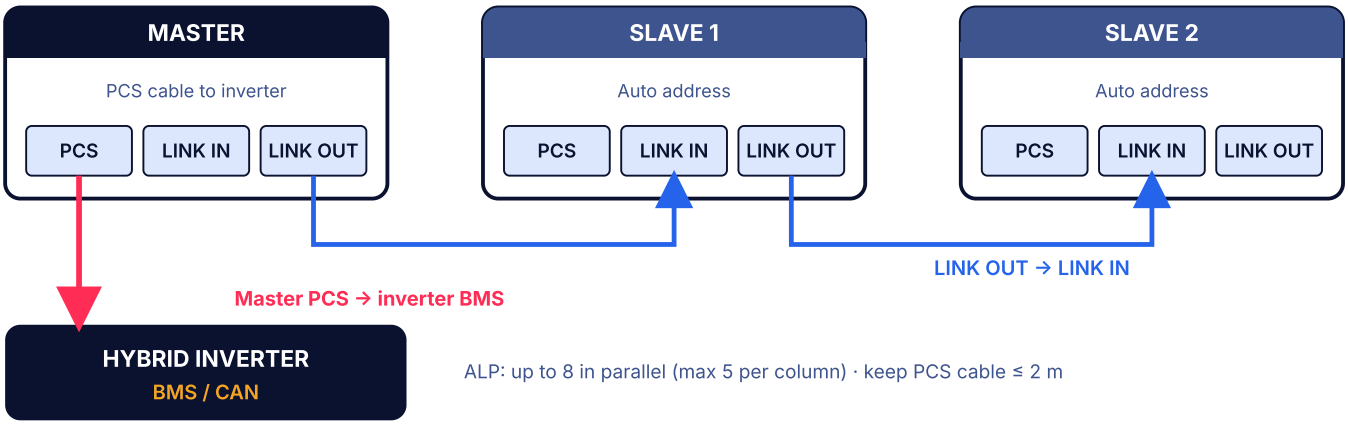
ALP 5.0L-E1 low-voltage battery (51.2 V) · notes for ARK LV and AXE 5.0L stacks

One rule for every brand: only the **master** battery talks to the inverter. All other batteries are chained to it.

1 · Set the master / address

No DIP switch and no master setting. Growatt LV batteries address themselves over the internal CAN link. The battery whose **PCS** port is cabled to the inverter acts as the head (master) unit. All packs must be the same model, cell type and firmware.

2 · Connect the communication cables



1. Inverter **BMS** port → **PCS** port of the master (head) pack.
2. **Master LINK OUT** → **Slave 1 LINK IN**, then Slave 1 LINK OUT → Slave 2 LINK IN, and so on.
3. PCS port pinout: CAN-H pin 4, CAN-L pin 5, GND pins 3 and 6, wake pins 7 and 8. Use a straight pin-to-pin RJ45 cable.
4. **ARK LV:** same idea, LINK OUT of the top module → LINK IN of the next module down. **AXE 5.0L:** packs in one cluster link through the stacking connector; LINK IN / OUT is only for joining clusters.

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3 · Inverter settings

- **SPF (off-grid):** Program 05 = **LI**, then Program 36 = **L51** (CAN). L01–L15 are RS485, L51–L65 are CAN; use only one.
- **SPH / SPM hybrid:** Setup → Mode Change → Battery Type = **Lithium**, protocol 01 over CAN.
- **Parallel SPF inverters:** only the first (L1) inverter takes the battery cable.
- Check SOC and pack count on the inverter after start-up.

4 · Power-on sequence

1. Connect all communication cables before any power.
2. Close each pack's breaker.
3. Hold the power button on one pack for 3–5 seconds (ARK: 2 seconds). One press wakes every pack in the chain.
4. Green RUN / ALM flashing together = parallel link OK.
5. Start the inverter. Packs switch off after about 25 minutes if the inverter does not communicate.

Install packs at a similar charge level (about 35–45% SOC) and add new packs within 1 year of the originals. Do not mix C1 and P1 AXE packs.

5 · Cable & good practice

Caps	Fit the dust caps on unused RJ45 and USB ports.
Cable	Standard straight LAN cable (CAT5e/CAT6, T568B both ends). No crossover.
Routing	Keep communication cables away from DC power cables. Plugs must click in fully.
Mixing	Never mix brands, models or capacities in one battery bank or one communication chain.
Voltage	Before paralleling, charge or rest all packs so their voltages are close.

6 · Troubleshooting: "no SOC" or "BMS communication fail"

"External CAN fail"	Check the PCS cable between the master and the inverter.
"Interior CAN fail"	Check the LINK OUT → LINK IN cables between packs.
"Type inconsistency of PACK"	Packs are different models or firmware. Use matching packs only.
SPF warning 20 / 33	BMS communication error or loss: check Program 36 (L51) and the cable.
Packs switch off by themselves	No inverter communication for 25 minutes: check the PCS cable and inverter settings.